Yuchen (David) Li

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Experience

- **Software Developer** Oct 2023 – Present
  Advantive LLC – formerly PINpoint Information Systems Inc. *Toronto, ON, Canada*
  - **Led development-side delivery** of a parts-lookup enhancement, resolving customer requirement gaps and designing a **dual-mode search workflow** across station-specific shortlists and large parts catalogs; implemented **.NET 10** desktop functionality and supported **Angular** delivery.
  - Improved a critical data-processing workflow by approximately **80–90% in the desktop application** and **60–70% in the web application** through cross-component profiling and bottleneck elimination, and cut hierarchy load times by **about 90%** by replacing **over 3,600 sequential server round trips** with a single bulk query.
  - Independently designed and delivered a configurable **.NET Windows service** automating third-party work-order submissions via **REST APIs**; owned file monitoring, retries, error recovery, logging, and archiving while evaluating evolving customer requests for feasibility and risk.
  - Modernized legacy applications and services to **.NET 8/10**, authored **Azure DevOps YAML CI/CD pipelines**, and restored previously skipped automated tests to release validation.
  - Remediated critical authentication vulnerabilities involving **JWT refresh failures** and **cross-user session isolation**; replaced legacy token validation and upgraded **OpenID Connect** dependencies.
  - Diagnosed silent data loss and reconnect storms in production **OPC UA integrations** through protocol analysis and controlled reproduction, delivering same-day mitigations and configurable fixes.
  - Resolved critical post-upgrade production incidents, including **thread-pool starvation** causing **2–10-minute station freezes**, delivering safe patches within one business day.
  - Served as **Scrum Master** alongside development, improving sprint planning and urgent-work visibility; mentored junior developers and reviewed code for concurrency and data-integrity risks.

- **Software Developer (Co-op)** Sep 2022 – Apr 2023
  PINpoint Information Systems Inc. *Burlington, ON, Canada*
  - Built a multi-threaded **database replay tool** that improved testing **throughput** by **10X**, significantly reducing QA bottlenecks and strengthening core system scalability.

- **Software Developer (Co-op)** May 2021 – Aug 2021
  Digital 14 *Mississauga, ON, Canada*
  - Developed secure **Java-based backend services** for communication platforms while supporting enterprise-grade code reviews and **CI/CD** security adoption.

- **Software Developer (Co-op)** Jan 2020 – Apr 2020
  JSI Telecom *Kanata, ON, Canada*
  - Contributed to the **UI/UX** development of core product modules, collaborating within an **Agile** team to deliver iterative enhancements for enterprise clients.

Technical Skills

Languages:	C#, SQL, TypeScript, JavaScript, Java
Frameworks:	.NET 8/10, ASP.NET Core, Angular, Entity Framework Core, WPF
Data:	SQL Server, T-SQL, Stored Procedures, Query Optimization, Indexing
Tools:	Azure DevOps (YAML Pipelines), Git, PowerShell, Postman, Visual Studio, IIS
Engineering:	REST APIs, OAuth 2.0, OpenID Connect, JWT, OPC UA (KEPServerEX), Concurrency, Performance Profiling, MVVM

Education

- **University of Waterloo** Waterloo, ON
  Bachelor of Computer Science – Co-operative Program (Honours) *Sep 2018 – Aug 2023*
  - **Scholarships:** President's Scholarship of Distinction